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Prof. Stephen T. C. Wong and Prof. Xiaoyin Xu
Co-Editors-in-Chief
Computerized Medical Imaging and Graphics
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Dear Editors-in-Chief,

We submit the manuscript “**Pseudo-3D Heightfield Reconstruction from Monocular Endoscopic Video via Gradient-Anchored Geometric Constraints**” as a Research Article for consideration in *Computerized Medical Imaging and Graphics* (CMIG).

Why CMIG. The paper addresses a computational and visualization problem in endoscopic imaging: recovering a metric, renderable 3D surface from monocular (and, when available, stereo) surgical video. Collapsed or planar reconstructions cannot support augmented-reality overlay or image-guided measurement. The proposed 2.5-D heightfield is both a geometric representation and a visualizable mesh, which matches CMIG’s scope in reconstruction, visualization, endoscopy, and image-guided intervention.

What is new.

- A gradient-anchored heightfield that preserves tissue-surface topology while suppressing specular artifacts, without learned natural-image priors.
- The same heightfield drives camera pose estimation and dense epipolar matching, so localization and reconstruction share one 3D signal.
- An optional calibrated stereo variant that recovers metric depth (AbsRel 0.0285 on EndoNeRF) while retaining the heightfield as a geometric regularizer.

Evidence. On an endoscopic chessboard benchmark, the proposed method and Stereo SGM are the only approaches that produce Normal 3D point clouds; seven feed-forward and SfM baselines collapse to cones or planes. Replacing heightfield-ICP poses with a 2D essential-matrix front-end collapses the cloud, isolating the heightfield signal. On SCARED, a 22-sequence stereo-laparoscopic set, and EndoNeRF, the method matches Stereo SGM in depth accuracy and supplies a confidence map for visualization.

Declarations. This manuscript is not under consideration elsewhere. All authors have approved the submission. The study uses publicly available or de-identified data and does not involve new human-subject or animal experiments. The authors have no competing interests. Research data and code will be deposited upon acceptance and are available from

the corresponding author in the meantime.

We thank you for considering this work.

Sincerely,

Ranyang Li, Junjun Pan, et al.